

14.2

18. **Three Entrances** A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$H_0: p_1 = p_2 = p_3 = \frac{1}{3}$ all entrances have same probability

H_a : at least one entrance proportion is different

$\alpha = 0.05$, $\alpha/3 = 0.0167$, $\alpha/3 = 0.0167$, $\alpha/3 = 0.0167$

$E_i = np_i$

$E_1 = 200(\frac{1}{3}) = 66.67$

$E_2 = 200(\frac{1}{3}) = 66.67$

$E_3 = 200(\frac{1}{3}) = 66.67$

Test: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

$$\chi^2 = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67} = 5.99 + 0.50 + 1.61 = 8.10$$

3 degrees

$d.f. = 3 - 1 = 2$

$\alpha = 0.05 \rightarrow \chi^2_{\alpha, 2} = 5.991$

$R.R. \chi^2 > 5.991$

$8.10 > 5.991$ (Reject H_0)

→ There is sufficient evidence at 0.05 significance level to conclude that the three entrances are not equally preferred

95% CI for entrance 1 proportion

$$\hat{p} = \frac{83}{200} = 0.415$$

$$CI: \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.415 \pm 1.96 \sqrt{\frac{0.415(0.585)}{200}}$$

$$0.415 \pm 0.0605$$

$$(0.3545, 0.4755)$$

14.3 - Two Categorical Variables

18. **Fatal Accidents** Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large	Total
Fatal	67	26	16	109
Not Fatal	128	63	46	237
Total	195	89	62	346

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobile? Test using a 5% significance level.

H_0 : frequency and automobile size are independent
 H_a : frequency and automobile size are dependent

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{n}$$

$$(109) E_{11} = \frac{109 \times 195}{346} = 61.95 \quad (109) E_{12} = \frac{109 \times 89}{346} = 27.97 \quad (109) E_{13} = \frac{109 \times 62}{346} = 19.57$$

$$(237) E_{21} = \frac{237 \times 195}{346} = 133.05 \quad (237) E_{22} = \frac{237 \times 89}{346} = 61.03 \quad (237) E_{23} = \frac{237 \times 62}{346} = 42.93$$

$$(195) E_{31} = \frac{195 \times 109}{346} = 61.03 \quad (195) E_{32} = \frac{195 \times 89}{346} = 50.07 \quad (195) E_{33} = \frac{195 \times 62}{346} = 34.90$$

$$\text{Test: } \chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \frac{(67-61.95)^2}{61.95} + \frac{(26-27.97)^2}{27.97} + \frac{(16-19.57)^2}{19.57} + \frac{(128-133.05)^2}{133.05} + \frac{(63-61.03)^2}{61.03} + \frac{(46-42.93)^2}{42.93}$$

$$= 0.505 + 0.147 + 0.650 + 0.352 + 0.066 + 0.195 = 1.815$$

$$d.f. = (3-1)(3-1) = 2$$

$$\alpha = 0.05 \rightarrow \chi^2_{\alpha, 2} = 5.991$$

$$R.R. \chi^2 > 5.991$$

$$1.815 < 5.991$$
 (Fail to reject H_0)

→ There is no sufficient evidence at 0.05 significance level to conclude that the frequency of fatal accidents depends on automobile size

22. **Heart Attacks on Mondays** Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day. In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	36
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$H_0: p_{Sun} = p_{Mon} = p_{Tue} = p_{Wed} = p_{Thu} = p_{Fri} = p_{Sat} = \frac{1}{7}$
 H_a : at least one day has a different proportion

$$E_i = n \left(\frac{1}{7} \right) = \frac{200}{7} = 28.571$$

$$\text{Test: } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(24-28.571)^2}{28.571} + \frac{(36-28.571)^2}{28.571} + \frac{(27-28.571)^2}{28.571} + \frac{(26-28.571)^2}{28.571} + \frac{(32-28.571)^2}{28.571} + \frac{(36-28.571)^2}{28.571} + \frac{(29-28.571)^2}{28.571}$$

$$= 0.791 + 1.892 + 0.086 + 0.291 + 0.412 + 0.291 + 0.006 = 3.669$$

$$d.f. = 7 - 1 = 6$$

$$\alpha = 0.05 \rightarrow \chi^2_{\alpha, 6} = 12.592$$

$$R.R. \chi^2 > 12.592$$

$$3.669 < 12.592$$
 (Fail to reject H_0)

→ There is no sufficient evidence at 0.05 significance level to conclude that the incidence of heart attacks differs by day of the week.

23. **Telecommuting** As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 500 workers were classified according to salary and number of workdays per week spent at home.

Salary	Workdays at Home per Week		
	Less Than One	At Least One, Not More Than All	All at Home
Under \$10,000	18	16	14
\$10,000-\$14,999	14	20	12
\$15,000-\$19,999	15	22	9
Above \$19,999	13	20	12
Total	60	78	46

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the p-value for this test of hypothesis. Does the p-value confirm your conclusion from part a?

a) H_0 : salary and number of workdays at home are independent
 H_a : salary and number of workdays at home are dependent

Expected table

	less than 1	at least 1	all at home
< \$10k	9.60	21.60	18.80
\$10k-14.99k	9.60	28.80	14.40
\$15k-19.99k	9.60	28.80	11.60
> 20k	9.60	22.40	11.60

$$\text{Test: } \chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 5.99$$

$$d.f. = (4-1)(3-1) = (3-1)(3-1) = 2$$

$$\alpha = 0.05 \rightarrow \chi^2_{\alpha, 2} = 5.992$$

$$R.R. \chi^2 > 5.992$$

$$5.99 < 5.992$$
 (Fail to reject H_0)

→ There is no sufficient evidence at 0.05 significance level to conclude that salary is dependent on number of workdays spent at home

$$H_0: p_1 = p_2 = p_3 = \dots = p_k = \frac{1}{k}$$

$$H_a: \text{at least one } p_i \neq \frac{1}{k}$$

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$E_i = n \left(\frac{1}{k} \right) = \frac{n}{k}$$

$$d.f. = k - 1$$

$$R.R. \chi^2 > \chi^2_{\alpha, k-1}$$

$$\text{Confidence Interval}$$

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$b) d.f. = 6$$

$$\chi^2 = 5.99$$

$$p = 0.98 \text{ since } p > 0.05$$

→ Yes, the p-value confirms the conclusion that we do not reject H_0 .